

Markus Zetto

Curriculum Vitæ

Office Address: Room 4.32
Bundesstraße 55
20146 Hamburg
Germany

* 22 July 2000, German

+49 40 42838 5305

✉ markus.zetto@uni-hamburg.de

🌐 www.markus-zetto.com



Education

- 2023–now **PhD in Mathematics, Hamburg University**
with Dr. David Reutter at his Emmy Noether research group
"Topological quantum field theory beyond three dimensions"
Topic: *Higher fusion categories* - Enriched ∞ -categories and applications to extended topological field theories
- 2021–2023 **Master in Mathematics, Heidelberg University, 1.0**
Thesis: *L-groups of Sheaves on Stratified Spaces* supervised by Prof. Dr. Markus Banagl
- 2020–2023 **Master in Physics, Heidelberg University, 1.0**
Thesis: *Constructible Factorization Algebras for Field Theories* supervised by Prof. Dr. Johannes Walcher & Prof. Dr. Markus Banagl
- 2017–2020 **Bachelor in Physics, Heidelberg University, 1.0**
Thesis: *Non-affine Landau-Ginzburg Models and Matrix Factorizations* supervised by Prof. Dr. Johannes Walcher
- 2009–2017 **Abitur, Leibniz-Gymnasium, Rottweil, 1.0**

Preprints and Publications

- planned 2026 **Cauchy-complete (∞, n) -categories and higher idempotents**, with David Reutter
- planned 2025 **Enriched ∞ -operads and symmetric sequences**, draft available at www.markus-zetto.com/enriched-operads.pdf
- January 2025 **Enriched ∞ -categories as marked module categories**, with David Reutter, <https://arxiv.org/abs/2501.07697>

Scholarships and Prizes

- 2017–2023 Scholarship by "Studienstiftung des deutschen Volkes"
- 2019–2021 Participation in the yearly, international PLANCKS physics team contest
- 2019–2022 Participation in the yearly, national DOPPLERS physics team contest
- 2017 Silver Medal at the International Physics Olympiad in Indonesia
- 2017 Silver Medal at the European Physics Olympiad in Estonia

Futher Activities

- Co-organization of a *Derived Algebraic Geometry* online reading seminar 2025
- Supervision & talks at, and general engagement in gifted youth programs, like the German selection competition for the International Physics Olympiad 2020-2022 and 2024, the Orpheus-Seminar twice in 2021 and once in 2024, and "Tag der Talente" 2021

Research Interests

- Enriched ∞ -categories and (∞, n) -categories
- Extended topological field theories
- Gray categories and categorical spectra
- Derived geometry, factorization algebras and Lagrangian field theory
- Stratified homotopy theory and topology
- Algebraic L-theory and applications in high-dimensional topology

Invited Talks

- upcoming **European Quantum Algebra Lectures (EQuAL)**, Online, "Cauchy-completions and Higher idempotents"
- 03.12.25
- 24.10.25 **UCLouvain**, "Cauchy completions and higher category Theory"
- 17.04.25 **Göttingen (Higher Structure Seminar)**, "Absolute Colimits and Extended TFTs "
- 18.12.24 **Vienna (Mathematical Physics Seminar)**, "Higher vector spaces"
- 13.03.24 **Vienna (Mathematical Physics Seminar)**, Online, "Cauchy completions and higher idempotents"

Contributed and selected Seminar Talks

- 09.10.25 **Derived Algebraic Geometry Seminar**, Online, "Derived Artin Stacks"
- 29.06.25 **Kolding (European Talbot)**, "Operads and Koszul duality"
- 19.06.25 **Hamburg (ZMP Seminar)**, "Introduction to cobordism theory"
- 25.02.25 **Raitenhaslach (Workshop on Gray Products and Lax Constructions)**, Model-Independent approach to the Gray product of $(\infty, 2)$ -categories
- 27.01.25 **Hamburg (Topology Seminar)**, Deformation Theory via formal moduli problems
- 13.12.24 **Hamburg (Quantum Topology Seminar)**, "Higher vector spaces"
- 23.09.24 **Glenmore (Scottish Talbot on Algebra and Topology)**, "Drinfeld centers of fusion 2-categories"
- 03.06.24 **Hamburg (Higher Structures Seminar)**, " Cauchy-complete ∞ -categories and lax additivity"
- 18.04.24 **Utrecht ((∞, n) -categories and their applications)**, "Cauchy completions and higher category Theory"

- 12.12.23, **Hamburg (Research Seminar Algebra and Mathematical Physics)**, Lecture
 19.12.23, Series on ∞ -operads, stable ∞ -categories and factorization homology
 09.01.24,
 30.01.24
 20.11.23 **Hamburg (Higher Structures Seminar)**, "L-Groups of higher sheaves"
 20.09.23 **Kiel (Conference on Characteristic Classes and Singular Spaces)**, "L-Groups of stratified spaces"

Visited Conferences and Workshops

- 11.-15.08.25 Masterclass: Infinity Operads and Applications to Geometry (Kopenhagen, Denmark)
 20.-27.07.25 Conference: Spaces of Tensor Categories (Edinburgh, Great Britain)
 29.06.-05.07.25 European Talbot - Higher algebra and chromatic homotopy theory (Kolding, Denmark)
 24.-27.02.25 Workshop on Gray Products and Lax Constructions (Raitenhaslach, Germany)
 16.-21.02.25 Cobordism categories Masterclass: Building bridges between algebra and geometry (Copenhagen, Denmark)
 09.-13.09.24 Workshop on "Dualisable Categories & Continuous K-theory" (Bonn, Germany)
 22.-28.09.24 Scottish Talbot On Algebra and Topology - Higher tensor categories and their extensions (Glenmore, Great Britain)
 10.-14.06.24 Masterclass: Continuous K-theory, dualizable and rigid categories (Copenhagen, Denmark)
 15.-19.04.24 (∞, n) -categories and their applications (Utrecht, Netherlands)
 08.-09.04.24 Bavarian Geometry & Topology Meeting XII (Regensburg, Germany)
 18.-22.09.23 Conference on Characteristic Classes and Singular Spaces (Kiel, Germany)
 14.-18.08.23 Higher Structures in Functorial Field Theory (Regensburg, Germany)
 19.-23.06.23 Hausdorff School: TQFTs and their connections to representation theory and mathematical physics (Bonn, Germany)
 22.-24.03.23 Higher categorical methods in algebra and geometry (Hamburg, Germany)
 18.-22.07.22 Domoschool 2022: Gauge Theory and the Langlands Program (Domodossola, Italy)
 04.-08.10.21 Pure Spinors, Superalgebras, and Holomorphic Twists (Heidelberg, Germany)
 12.-16.07.21 Junior Researchers Conference: Higher Structures in QFT and String Theory (Online)

Languages

German	Native speaker
English	Fluent
Romanian	Basics
French	Basics (DELF B1)
Latin	Latinum